

REAL INTERNAL TOOLS. OPEN SOURCE.

7 repos from the most expensive engineering teams in the world.

Jane Street. Goldman. JPMorgan. BlackRock. HRT. Two Sigma. D.E. Shaw.

#1

Jane Street

[janestreet/magic-trace](#)

5.3k stars

**When your profiler
is blind,
this sees every CPU
instruction.**

Powered by Intel PT. Not sampling. Every instruction, in order, with exact timing.

Goldman Sachs

`goldmansachs/gs-quant`

MIT licensed

The derivative pricing library their traders use at their desks.

Not a textbook. Working code. MIT licensed. You can run it today.

#3

JPMorgan

[finos/perspective](#)

Bloomberg Terminal = \$24K/yr

What JPM traders use to watch markets in real time. Free.

Streaming data visualization. Handles tick-level data at scale. Maintained under FINOS.

#4

BlackRock

`blackrock/lcso`

Rust

Where scipy gives up, this works.

Layered Conic Solver for Optimization. Portfolio problems at scale. BlackRock manages \$10 trillion.

#5

Hudson River Trading

`hudson-trading/corral`

C++20

The foundation of HFT infrastructure at one of the largest U.S. trading firms.

Structured concurrency. Concurrent operations in a tree. Cancellation and error propagation that actually makes sense.

#6

Two Sigma

`twosigma/flint`

Apache Spark

Time-series joins when timestamps never match exactly.

Tick data from different exchanges, different clocks.
Temporal tolerance. Built for billions of ticks.

#7

D.E. Shaw

[deshaw/pyflyby](#)

Also funded IPython

Auto-import for Jupyter. But read the footnote.

D.E. Shaw funded IPython development. Every data scientist working in notebooks is using infrastructure they financed.

The algorithm is not the moat.

The infrastructure that runs it reliably and fast enough is the moat. That's what every one of these repos shows.

All 7 are public on GitHub.

The README files alone are worth reading. Full breakdown at bionicbanker.tech

Follow for more from inside finance and AI.